

Notes: Kovak (AER, 2013)

Regional Effects of Trade Reform: What is the Correct Measure of Liberalization?

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1 Big picture

- **Question.** How should researchers measure the exposure of local labor markets to national trade liberalization? The paper asks what regional treatment variable is implied by trade theory, and then tests it using Brazil's early-1990s tariff reform.
- **Core issue.** A national tariff reform changes prices at the industry level, but workers live in regional labor markets with different industry mixes. The local effect depends on which industries matter most in each region.
- **Theoretical framework.** The paper builds a specific-factors model of regional economies. Labor is mobile across industries inside a region but immobile across regions; the other input is industry-region specific.
- **Main theoretical result.** Regional wage changes should be a weighted average of liberalization-induced industry price changes. The weights depend on each industry's employment share in the region and its labor-demand elasticity.
- **Key measurement implication.** The relevant regional liberalization measure is not a simple average of tariff cuts. It is a local-industry-share weighted average of changes in $\log(1 + t_i)$, where t_i is the industry tariff rate.
- **Nontraded-sector insight.** The model shows that nontraded prices move with traded-goods prices during liberalization. Therefore, when constructing the regional tariff-change measure, the preferred approach is to omit the nontraded sector and rescale the traded-sector weights.
- **Empirical setting.** The paper studies Brazil's trade liberalization from 1990 to 1995 and regional wage changes from the 1991 and 2000 Brazilian demographic censuses.
- **Main empirical result.** Regions whose workers were concentrated in industries with larger tariff cuts experienced larger wage declines. A region facing a 10 percentage point larger liberalization-induced price decline experienced roughly a 4 percentage point larger wage decline or smaller wage increase.
- **Magnitude discipline.** The preferred estimates are between 0 and 1, as the model predicts when pass-through and interregional migration are incomplete.
- **Economic takeaway.** Trade reform has local distributional consequences. Correctly measuring local exposure requires a theory-based regional price-shock index, and mishandling nontraded sectors can severely distort estimated effects.

2 Incremental contribution of the paper

- **From intuitive shift-share exposure to theory-based exposure.** Earlier regional trade papers often used local industry shares to weight national trade shocks. This paper gives that practice a formal foundation using a regional specific-factors model.
- **Correct regional treatment construction.** The paper shows exactly what the local liberalization measure should contain: industry price changes, local employment weights, and labor-demand elasticity adjustments.

- **Treatment of nontradables.** A major contribution is the nontraded-sector result. The model argues against assigning a zero price change to nontradables when studying wages, because nontraded prices endogenously move with traded prices.
- **Quantitative warning for empirical work.** The paper demonstrates that setting the nontraded-sector price change to zero inflates the estimated wage effect by more than four times in the Brazilian application.
- **Application to a canonical reform.** Brazil's liberalization is useful because tariff reductions varied sharply across industries, and Brazilian microregions had different pre-reform employment structures.
- **Theoretical guidance on weights.** The paper clarifies why industries with larger regional employment shares and more elastic labor demand should receive more weight.
- **Link to local-labor-market designs.** The result is not limited to tariff liberalization. The same logic applies to any national industry-level price shock that affects local labor demand through regional industry mix.
- **Empirical validation of the model.** The wage regressions produce positive coefficients of the expected magnitude: local wages fall more where liberalization-induced price declines are larger.
- **Policy relevance.** The paper helps interpret trade liberalization as a policy that creates spatially uneven wage effects, not merely national average effects.

3 Data and measurement

3.1 Brazilian trade liberalization

- Brazil pursued import-substituting industrialization for much of the twentieth century, using high tariffs and nontariff barriers to protect domestic firms.
- Before liberalization, tariff levels were high and varied substantially across industries. In 1987 the average tariff was about 54.9%, ranging from 15.6% for oil, natural gas, and coal to 102.7% for apparel.
- Major reforms began in March 1990, when Brazil removed important nontariff barriers and special customs regimes and converted protection into tariffs through tariffization.
- Between 1991 and 1994 Brazil reduced tariff levels and tariff dispersion across industries. After 1994 there was a small reversal, but the tariff structure remained broadly stable.
- The paper measures liberalization using the 1990-1995 proportional change in one plus the tariff rate, $d \ln(1 + t_i)$.

Interpretation: The 1990 tariff structure is treated as the starting point because tariffs had become the main protection instrument after the removal of major nontariff barriers.

3.2 Industry tariff data

- Trade policy data come from Brazilian Applied Economics Research Institute (IPEA) sources, especially Kume, Piani, and de Souza’s tariff data.
- Tariff data are measured at the Nível 50 industry classification, similar to two-digit SIC.
- The preferred measure is nominal tariffs, though the author reports that effective rates of protection produce similar conclusions.
- For the final analysis, the data are harmonized into 21 industries, including agriculture and nontraded goods.

Interpretation: The paper needs industry tariff changes that are comparable to the industry categories in the population census, so crosswalk construction is a central data step.

3.3 Regional wage and employment data

- Wage and employment data come primarily from the 1991 and 2000 Brazilian Demographic Censuses.
- Local labor markets are microregions: groups of economically integrated contiguous municipalities with similar geographic and productive characteristics.
- The original 558 microregions are aggregated to 494 consistent microregions to account for boundary changes between 1991 and 2000.
- The wage sample focuses on individuals aged 18 to 55 who are not currently enrolled in school; wage regressions use workers with positive wage income.
- Wages are measured as earnings divided by hours.
- 1991 employment shares by industry are used to measure pre-liberalization regional industry composition.

Interpretation: Using 1991 as the baseline is not ideal because liberalization began in 1990, but the author argues that wages and employment shares adjusted slowly enough that 1991 still captures pre-adjustment regional exposure.

3.4 Regional wage changes

The paper estimates regional wage changes by running separate wage regressions in 1991 and 2000:

$$\log(w_{irt}) = X'_{irt}\Gamma_t + \text{industry fixed effects} + \text{microregion fixed effects} + u_{irt}. \quad (1)$$

- The controls remove changes in observable worker characteristics and changing returns to those characteristics.
- The microregion fixed effects are then normalized relative to the average log wage change.
- The resulting dependent variable is the adjusted 1991-2000 wage change in each microregion.

Interpretation: The wage outcome is a regional residual wage change, not a raw average wage change. This focuses the analysis on local wage movements net of worker-composition shifts.

3.5 Region-level tariff change

The core treatment variable is the region-level tariff change, denoted RTC_r :

$$RTC_r = \sum_{i \neq N} \beta_{ri} d \ln(1 + t_i), \quad (2)$$

where the preferred empirical weights are

$$\beta_{ri} = \frac{\lambda_{ri} \frac{1}{\theta_i}}{\sum_{i' \neq N} \lambda_{ri'} \frac{1}{\theta_{i'}}}. \quad (3)$$

- λ_{ri} is the fraction of region r 's labor force in industry i .
- θ_i is the industry-specific factor's cost share, calculated as one minus the wage-bill share of industry value added.
- The nontraded sector N is omitted and traded-sector weights are rescaled to sum to one.

Interpretation: A more negative RTC_r means the region was more exposed to industries whose tariffs and therefore producer prices fell more during liberalization.

4 Models and empirical specifications

4.1 Specific-factors model of regional economies

Each region is modeled as a specific-factors economy. Labor moves across industries inside the region, but not across regions. The second input is fixed to an industry and region.

The model's main wage equation is

$$\hat{w}_r = \sum_i \beta_{ri} \hat{P}_i, \quad (4)$$

where hats denote proportional changes.

- $\hat{P}_i$ is the national price change in industry i .
- $\beta_{ri} > 0$ and the weights sum to one across industries in each region.
- The weights are larger when an industry employs a larger share of regional labor or has a more elastic labor demand curve.

Interpretation: All regions face the same national industry price shocks, but regions differ in how much those shocks matter because their employment structures differ.

4.2 Labor-demand elasticity and weights

The theoretical weight depends on regional employment shares and on the elasticity of labor demand holding the specific factor fixed. In the empirical implementation, the author assumes Cobb-Douglas production, which implies

$$\sigma_{ri} = 1, \quad \theta_{ri} = \theta_i. \quad (5)$$

- This yields tractable weights using employment shares and national industry factor shares.
- Because detailed region-by-industry labor-demand elasticities are unavailable, this is a practical approximation.
- The paper also tests a version without the labor-share adjustment and finds very similar results in Brazil.

Interpretation: The labor-demand adjustment is theoretically correct, but empirically it has little effect here because adjusted and unadjusted industry weights are highly correlated.

4.3 Nontraded sector model

The paper extends the model to include a nontraded sector. It shows that the proportional change in the nontraded price is itself a weighted average of traded goods' price changes:

$$\hat{P}_{rN} = \sum_{i \neq N} \xi_{ri} \hat{P}_i. \quad (6)$$

- Nontraded prices do not remain fixed when traded prices change.
- If traded prices fall, local wages and local demand adjust, moving nontraded prices as well.
- Therefore, setting the nontraded sector's price change to zero is inconsistent with the model for wage outcomes.

Interpretation: The correct empirical treatment is to omit the nontraded sector from the weighted average and rescale traded-sector weights, rather than include nontradables with zero tariff change.

4.4 Wage-tariff estimating equation

The main empirical regression is

$$d \ln(w_r) = \zeta_0 + \zeta_1 RTC_r + \varepsilon_r. \quad (7)$$

- $d \ln(w_r)$ is the adjusted 1991-2000 wage change in microregion r .
- RTC_r is the region-level tariff change from 1990 to 1995.
- Regressions are weighted by the inverse of the squared standard error of the estimated regional wage change.
- Standard errors are clustered at the state level.

- Specifications are reported with and without state fixed effects.

Interpretation: The coefficient ζ_1 measures whether regions more exposed to tariff-induced price declines experienced larger wage declines.

4.5 Expected magnitude

The model suggests ζ_1 should be positive and plausibly between zero and one.

- If local labor markets were completely immobile and tariff changes fully passed through to producer prices, the coefficient could be close to one.
- Interregional migration smooths local wage effects and pushes the coefficient toward zero.
- Imperfect tariff pass-through and measurement error also reduce the coefficient.

Interpretation: A positive coefficient between zero and one is exactly the kind of magnitude the model predicts.

5 Identification and empirical design

5.1 Source of identifying variation

- Identification comes from two interacting sources of variation: national industry-level tariff cuts and regional pre-reform employment shares.
- Brazil's tariff cuts differed sharply across industries because pre-reform tariffs differed sharply across industries.
- Brazilian microregions differed in their employment concentration across agriculture, apparel, metals, food processing, and other sectors.
- A region heavily concentrated in industries with large tariff cuts receives a more negative RTC_r .

Interpretation: This is a shift-share design: common industry-level shocks are shifted into local exposure using pre-existing regional industry shares.

5.2 Exogeneity of tariff changes

- The central identifying assumption is that industry tariff changes are unrelated to counterfactual regional wage growth, except through liberalization exposure.
- The paper argues that Brazilian tariff cuts were largely driven by a policy goal of reducing average tariff levels and compressing tariff dispersion across industries.
- Figure 1 shows that industries with higher pre-liberalization tariffs experienced larger cuts; the correlation between initial tariff levels and tariff changes is about -0.90 .
- Since the protective structure was set decades earlier, it is less likely that 1990-1995 tariff cuts were chosen to favor industries based on their contemporaneous counterfactual performance.

- The paper also notes that the tariff-change pattern would tend to bias against finding the main positive result if high-growth pre-liberalization industries received larger cuts.

Interpretation: The credibility of the design comes from the rule-like equalization of tariffs: sectors that had been historically highly protected got the largest reductions.

5.3 What the design identifies

- The design identifies relative effects across regional labor markets, not the national aggregate effect of liberalization.
- The coefficient compares wage changes in regions more exposed to tariff-induced price declines to regions less exposed.
- The estimates are reduced-form effects of tariff changes on regional wages, not a full structural estimate of all welfare effects of liberalization.
- The design is best suited to wage outcomes because the theory assumes full employment and adjustment through wages.

Interpretation: The paper measures how liberalization redistributed wage growth across space within Brazil.

5.4 Migration and adjustment

- Labor is assumed immobile across regions in the model, but Brazil has meaningful interregional migration.
- Over the 1991-2000 period, migration may have arbitrated away part of the initial regional wage shock.
- This means the empirical coefficient likely understates the short-run impact of the trade shock on local wages.
- The expected coefficient is therefore below one even when the theory's underlying channel is correct.

Interpretation: Migration is not simply a nuisance. It helps explain why the estimated wage effect is substantial but not one-for-one.

5.5 Limitations of the empirical design

- The paper does not directly observe regional producer price changes; tariff changes serve as reduced-form instruments for price changes.
- Detailed region-by-industry output and labor-demand elasticities are unavailable, so the weight construction requires Cobb-Douglas assumptions.
- The 1991 census is used as the baseline even though liberalization started in 1990.
- The model assumes a homogeneous labor force and full employment, so it does not directly analyze inequality, unemployment, poverty, or child labor.

- The analysis captures relative regional wage changes during a decade that may also contain other macroeconomic and regional shocks.

Interpretation: The paper is strongest as a theory-disciplined regional wage design. It is not intended to be a complete model of all labor-market and welfare consequences of liberalization.

6 Tables and figures

6.1 Figure 1: Relationship between tariff changes and preliberalization tariff levels

Read:

- Figure 1 plots each industry by its 1990 preliberalization tariff level on the horizontal axis and its 1990-1995 tariff change on the vertical axis.
- The negative slope is very strong: industries with high initial tariffs experienced the largest tariff cuts.
- The reported correlation is -0.899 , and the regression coefficient is -0.556 .
- Apparel and rubber had high initial protection and large tariff declines, while agriculture is close to zero or slightly positive.

Economic message: The figure supports the exogeneity argument. Liberalization largely compressed the inherited tariff structure rather than selectively cutting tariffs based on contemporaneous industry performance.

6.2 Figure 2: Regional wage changes

Read:

- Figure 2 maps adjusted regional wage changes across Brazilian microregions between 1991 and 2000.
- Lighter regions experienced larger wage declines relative to the national average, while darker regions experienced larger increases.
- The figure shows substantial spatial heterogeneity in wage growth.
- The outcome is not raw wage growth; it is based on normalized changes in microregion fixed effects from wage regressions.

Economic message: The map shows that Brazil's labor-market adjustment was spatially uneven. The empirical question is whether this spatial variation lines up with trade-shock exposure.

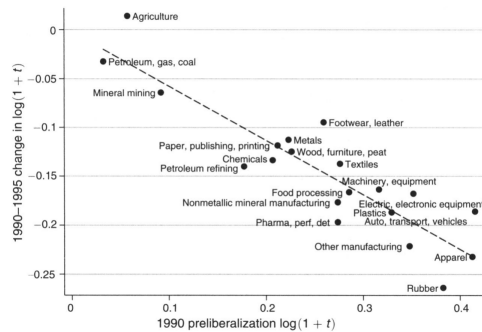


FIGURE 1. RELATIONSHIP BETWEEN TARIFF CHANGES AND PRELIBERALIZATION TARIFF LEVELS

Note: Correlation: -0.899 ; regression coefficient: -0.556 ; standard error: 0.064 ; t : -8.73 .

Source: Author's calculations based on data from Kume, Piani, and de Souza (2003).

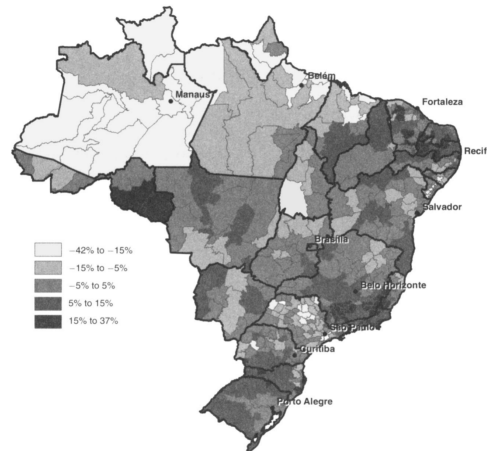


FIGURE 2. REGIONAL WAGE CHANGES

Note: Normalized change in microregion fixed effects from wage regressions.

(a) Figure 1: Tariff cuts were larger in initially more protected industries.

(b) Figure 2: Adjusted regional wage changes across Brazil.

6.3 Figure 3: Region-level tariff changes

Read:

- Figure 3 maps the theory-based regional tariff-change measure RTC_r .
- Lighter microregions faced larger tariff-induced price declines; darker regions faced smaller cuts or small increases.
- The measure varies across regions because regions differ in their pre-liberalization industry mix.
- Regions exposed to industries such as apparel, food processing, and other sectors with larger tariff cuts have more negative values.

Economic message: The map is the treatment variable. It translates one national reform into heterogeneous local shocks.

6.4 Figure 4: Demonstration of variation behind region-level tariff changes

Read:

- Figure 4 compares the industry weights for Rio de Janeiro and Traipu, Alagoas.
- Industries are sorted from the most negative to the most positive tariff changes.
- Rio de Janeiro places more weight on industries with large negative tariff changes, such as apparel and food processing.
- Traipu is almost entirely concentrated in agriculture, which had the most positive tariff change in the figure.

Economic message: The figure makes the shift-share logic transparent. Two regions face the same national industry tariff changes, but their exposure differs because their employment weights

differ.

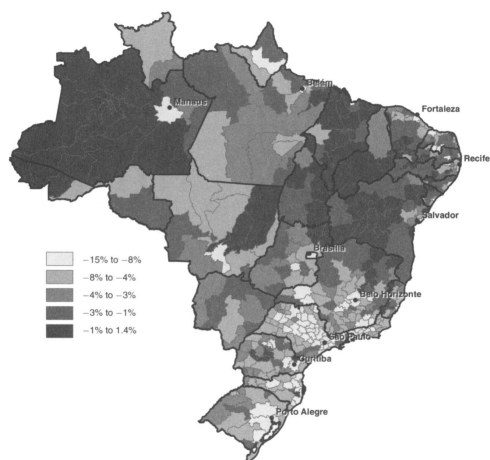


FIGURE 3. REGION-LEVEL TARIFF CHANGES

Notes: Weighted average of tariff changes. See text for details.

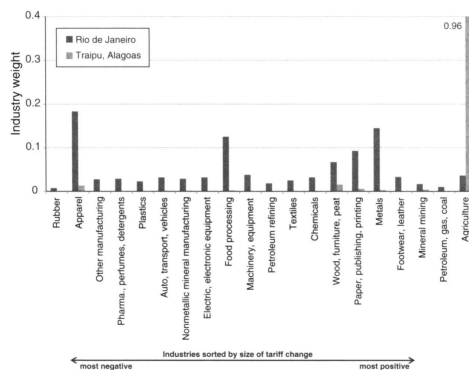


FIGURE 4. DEMONSTRATION OF VARIATION BEHIND REGION-LEVEL TARIFF CHANGES

Notes: Author's calculations. See text for details.

(a) Figure 3: Regional exposure to tariff-induced price changes.

(b) Figure 4: Why Rio de Janeiro and Traipu receive very different exposure measures.

6.5 Table 1: The effect of liberalization on local wages

Read:

- Columns (1) and (2) report the preferred specification. The coefficient on the regional tariff change is 0.404 without state fixed effects and 0.439 with state fixed effects.
- The positive sign means that more negative tariff shocks are associated with more negative wage changes. A 10 percentage point larger price decline implies roughly a 4 percentage point larger wage decline or smaller wage increase.
- Adding state indicators barely changes the point estimate but greatly improves precision; the estimate with state fixed effects is significant at the 1% level.
- Columns (3) and (4) omit the labor-share adjustment. The estimates remain almost identical, showing that this adjustment matters little in the Brazilian data.
- Columns (5) and (6) set the nontraded-sector price change to zero. The coefficients rise to 2.715 and 1.965, much larger than the theoretically plausible range.
- Columns (7) and (8) use nontraded-sector workers' wages as the dependent variable. The estimates are close to the preferred all-worker estimates, supporting the idea that nontraded wages move with traded-sector shocks.

Economic message: Table 1 is the core empirical result. The preferred regional tariff-change measure predicts local wage changes with the expected sign and magnitude, while the common alternative of setting nontradables to zero produces inflated estimates.

TABLE 1—THE EFFECT OF LIBERALIZATION ON LOCAL WAGES

	Main		No labor share adjustment		Nontraded price change set to zero		Nontraded sector workers' wages	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Regional tariff change	0.404	0.439	0.409	0.439	2.715	1.965	0.417	0.482
Standard error	(0.502)	(0.146)***	(0.475)	(0.136)***	(1.669)	(0.777)**	(0.497)	(0.140)***
State indicators (27)	—	X	—	X	—	X	—	X
Nontraded sector								
Omitted	X	X	X	X	—	—	X	X
Zero price change	—	—	—	—	X	X	—	—
Labor share adjustment	X	X	—	—	X	X	X	X
R ²	0.034	0.707	0.040	0.711	0.112	0.710	0.037	0.763

Notes: 493 microregion observations (Manaus omitted). Standard errors adjusted for 27 state clusters (in parentheses). Weighted by the inverse of the squared standard error of the estimated change in log microregion wage, calculated using the procedure in Haisken-DeNew, and Schmidt (1997).

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

Table 1: Main wage effects and alternative regional liberalization measures.

7 Short summary

- The paper asks how to correctly measure local exposure to national trade liberalization.
- A regional specific-factors model implies that wage changes should be a weighted average of industry price changes.
- The weights depend on local industry employment shares and industry labor-demand elasticities.
- The model implies that nontraded prices move with traded prices, so the nontraded sector should be omitted from the regional weighted average and traded weights should be rescaled.
- The empirical setting is Brazil's 1990-1995 trade liberalization, combined with 1991-2000 microregion wage changes.
- Industries with higher 1990 tariffs experienced larger tariff cuts, generating substantial cross-industry variation in price shocks.
- Regions with employment concentrated in industries facing larger tariff cuts experienced larger wage declines.
- The preferred estimate is about 0.4: a 10 percentage point larger liberalization-induced price decline predicts about a 4 percentage point larger regional wage decline or smaller wage increase.
- Omitting the labor-share adjustment has little impact in Brazil, but setting the nontraded-sector price change to zero inflates the estimates dramatically.
- The paper's main contribution is to give a theoretical foundation and practical guidance for regional trade-shock exposure measures.

8 Conclusion

8.1 Core conclusion

Kovak shows that regional exposure to trade liberalization should be measured using a theory-based weighted average of industry price changes. In Brazil, regions concentrated in industries

facing larger tariff cuts experienced larger wage declines. The evidence supports the specific-factors logic that national industry shocks translate into local wage effects through regional industry composition.

8.2 Economic intuition

The economic mechanism is simple. When tariffs fall in an industry, the price faced by domestic producers falls. That lowers labor demand in regions where the industry is important. Labor can move across industries inside the region, so the shock affects the region’s overall wage, not only the wage in the directly exposed industry. The size of the regional wage effect depends on how much regional labor was initially tied to the affected industries and how elastic labor demand is in those industries.

8.3 Incremental contribution revisited

The paper advances the regional trade literature by turning a common empirical practice into a model-derived object. It clarifies the correct weights, the interpretation of the coefficient, and the treatment of nontraded goods. This is especially important because local-labor-market papers often use shift-share exposure measures. Kovak shows when these measures are theoretically justified and when alternative constructions can be misleading.

8.4 Implications for empirical trade research

The paper implies that researchers should be careful when constructing local trade shocks. Industry shares should be measured before the shock, the industry shock should correspond to producer price changes, and nontraded sectors should not mechanically be assigned a zero price change when the outcome is wages. The result also suggests that national trade policy can create large regional distributional effects even when every region faces the same national tariff schedule.

8.5 Limitations

The analysis is reduced-form and does not directly estimate all producer-price changes, welfare effects, or general-equilibrium migration responses. The model assumes full employment and homogeneous labor, so it does not directly analyze unemployment, informality, poverty, or within-region inequality. The weight construction requires assumptions because detailed labor-demand elasticities and regional output by industry are unavailable. Finally, the 1991-2000 wage changes may reflect other regional shocks in addition to trade liberalization, although the theory-based treatment variable and the state fixed-effect specifications help discipline the interpretation.

8.6 Takeaway

The enduring takeaway is that the correct measure of regional liberalization is not just “how much tariffs fell” nationally. It is how much the industries important to a region were exposed to tariff-induced price changes. In Brazil, that measure strongly predicts regional wage outcomes, and the way researchers treat nontraded sectors can change the estimated magnitude dramatically.